**4. Program to Sort an Array using Selection Sort.** **Date:**

**AIM:**

To write the program to sort an array using insertion sort

**ALGORITHM:**

Step 1: start

Step 2: declare element count (n),

Step 3: get n

Step 4: declare arr [n]

Step 5: get arr [i]

Step 6: i++

Step 7: repeat from step 5 till i<n

Step 8: i = 0

Step 9 : minindex = i

Step 10: j = i+1

Step 11: if arr [i]<arr

Step 12: temp = arr [i]

arr[i] = arr[minindex]

arr[minindex] = temp

Step 13: j++

Step 14: repeat from step 11 till j<n

Step 15: i++

Step 16: repeat from step 9 till i<n-1

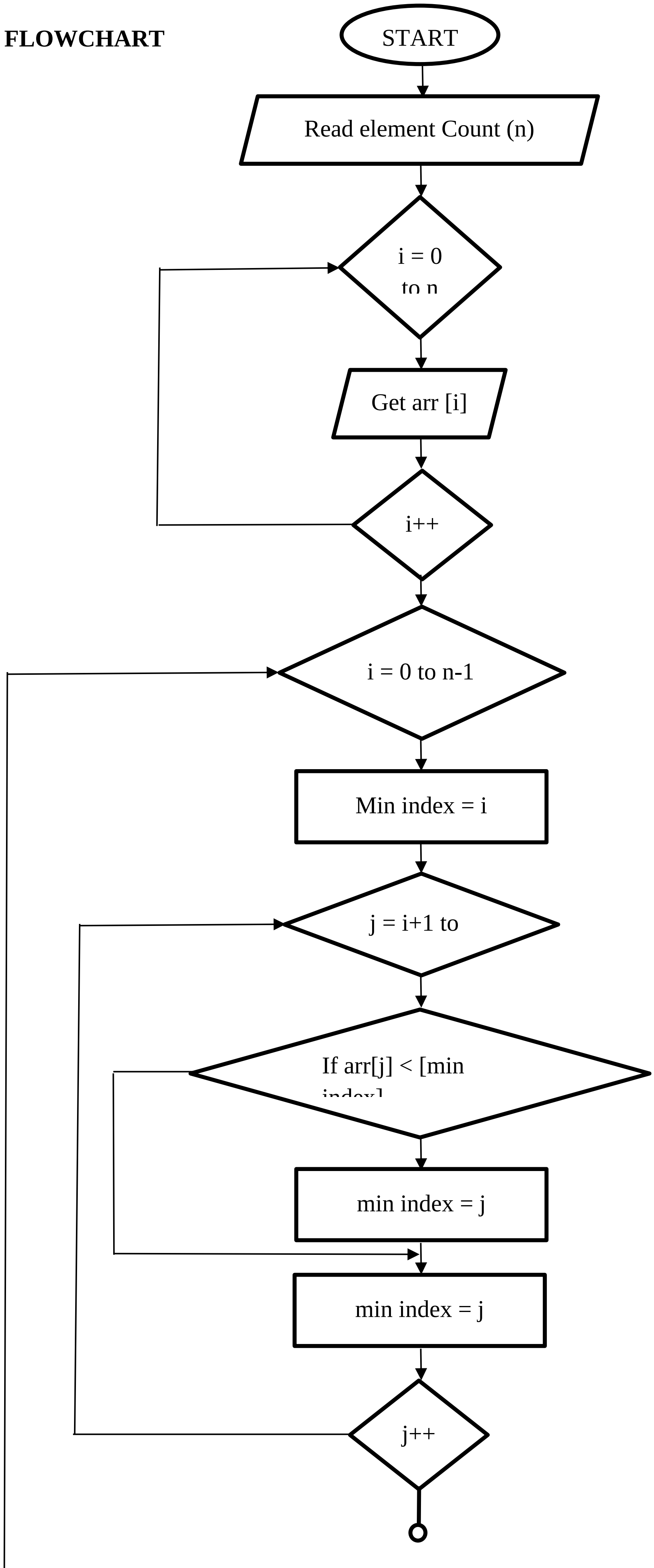
Step 17: stop i = 0

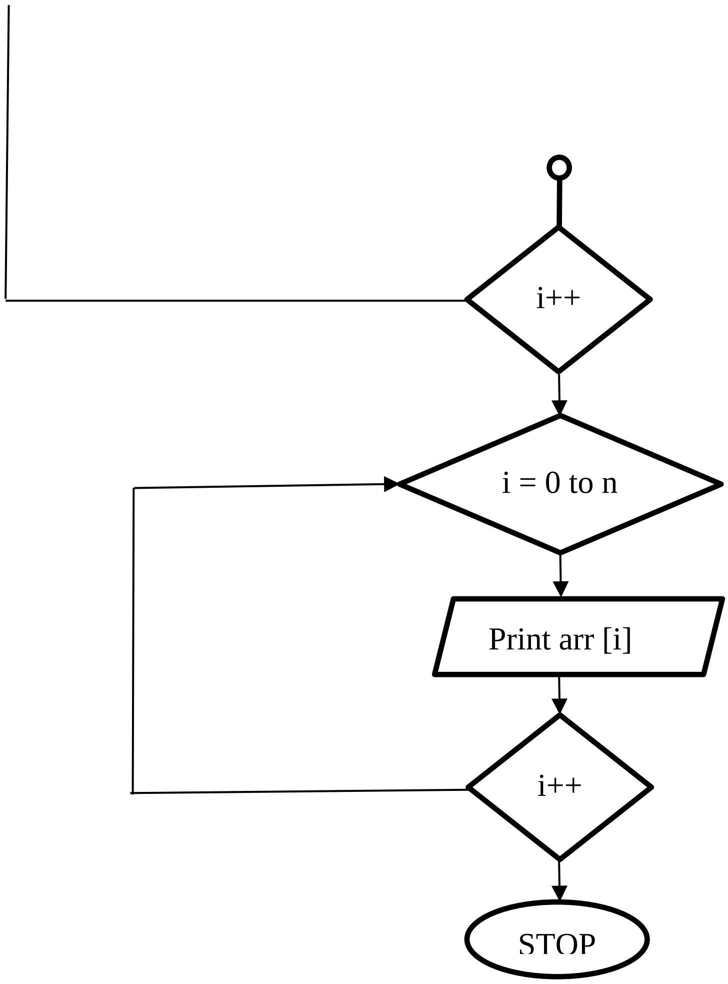
Step 18: print arr [i]

Step 19: i++

Step 20: repeat from step 18 till i<n

**FLOWCHART**


START
Read element Count (n)
i = 0
to n
Get arr [i]
i++
i = 0 to n-1
Min index = i
j = i+1 to
If arr[j] < [min
min index = j
min index = j
j++




**PROGRAM**

```
#include <stdio.h>
int main()
{
    int n, temp, minIndex;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter %d elements:\n", n);
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    // Selection Sort logic
    for (int i = 0; i < n - 1; i++)
```

```
    {
        minIndex = i;

        for (int j = i + 1; j < n; j++)
        {
            if (arr[j] < arr[minIndex])
                minIndex = j;
        }

        // Swap
        temp = arr[i];
        arr[i] = arr[minIndex];
        arr[minIndex] = temp;
    }

    printf("Sorted array:\n");
    for (int i = 0; i < n; i++)
        printf("%d ", arr[i]);

    return 0;
}
```

**SAMPLE I/O**

*Enter number of elements: 6*

*Enter 6 elements:*

*12*

*14*

*11*

*22*

*20*

*19*

*Sorted array:*

*11 12 14 19 20 22*